

Executive Analysis

Supply Chain Performance Analysis Using Data Analytics

1. Executive Overview

This project applies data analytics to an APL Logistics supply-chain dataset containing more than 180,000 transaction records. The analysis evaluates revenue, profitability, customer value, product and category performance, and discount-related profitability patterns.

An interactive Streamlit dashboard was developed to allow decision-makers to upload the cleaned dataset and explore the results through a set of focused analytical modules.

2. Business Purpose

The purpose of the analysis is to convert a large transactional dataset into practical information that can support business decision-making. The project focuses on understanding where revenue and profit are generated, how customers and segments contribute, which products and categories perform strongly or weakly, and how discount levels relate to profitability.

3. Key Findings

Revenue and Profitability — Revenue and profitability are not necessarily distributed equally. High sales do not automatically imply the strongest profit margin, so both measures should be monitored.

Customer Value — Customer contribution varies. Customer-level profitability and segment contribution provide a more useful view of commercial value than transaction volume alone.

Product and Category Performance — Products and categories should be assessed using revenue, profit, quantity, discounts, and margins. High-revenue/low-profit categories are important candidates for further review.

Discount Impact — Discount levels can be compared with profit and margin using discount bands and correlation analysis. The results indicate association rather than causation and should therefore be used as a decision-support signal rather than proof of a pricing effect.

4. Strategic Implications

The analysis supports a more balanced approach to supply-chain performance monitoring. Decision-makers should consider revenue, profitability, customer value, product performance, and discounting together rather than relying on a single KPI.

The dashboard also makes it possible to examine different markets, regions, customer segments, products, categories, and discount scenarios interactively.

5. Recommendations

1. Investigate high-revenue/low-profit categories for pricing, discount, cost, and operational issues.
2. Monitor customer profitability and identify high-value customer segments.
3. Evaluate products using both commercial scale and margin.
4. Review discount decisions for their potential profitability implications.
5. Continue using interactive analytics to support faster and more transparent business review.

6. Dashboard Deliverable

The Streamlit application contains four modules:

- Revenue & Profit Overview
- Customer Value Dashboard
- Product & Category Performance
- Discount Impact Analyzer

The application is deployed at <https://supplychainanalysis.streamlit.app/> and uses an upload-based workflow so the large dataset does not need to be stored in the repository.

7. Executive Takeaway

The central takeaway is that supply-chain performance should be evaluated using a combination of revenue, profitability, customer value, product/category performance, and discount behavior.

The project provides a practical framework for turning transactional supply-chain data into an interactive decision-support resource. Broader operational or policy conclusions would require additional external data, but the current analysis provides a solid foundation for business-level performance monitoring.

8. Limitations

The dataset does not provide a suitable date field for reliable monthly trend analysis. Correlation analysis does not establish causation, and the discount scenario is an analytical estimate rather than a financial forecast. External cost, infrastructure, economic, and supplier variables are outside the current dataset unless explicitly represented.

9. Conclusion

The combination of EDA, business interpretation, and interactive visualization provides a practical approach to supply-chain performance analysis. The completed dashboard, research paper, and executive analysis together form a concise and reusable analytics project.